Mars IoT Automation Platform

Distributed sensor ingestion, event-driven automation and real-time habitat monitoring — from Mars.

5
Microservices

15
Sensor Sources

4
Actuators

Mission Overview

Year 2036. You've been accidentally shipped to Mars while sleeping at your desk. The habitat automation stack is partially destroyed. Devices speak incompatible dialects — some stream telemetry, others respond only to polling. Your mission: rebuild the platform before thermodynamic consequences catch up.

Ingest

Poll REST sensors & consume telemetry streams from 15 heterogeneous devices

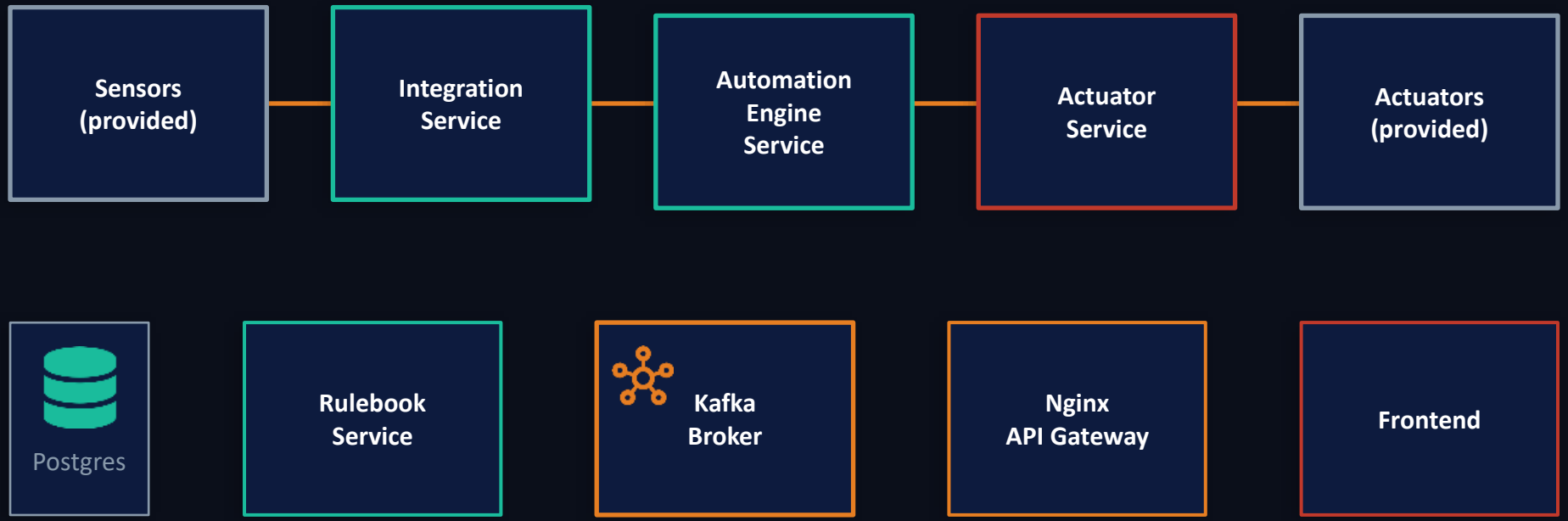
Automate

Evaluate IF-THEN rules on every incoming event and trigger actuator commands in real time

Monitor

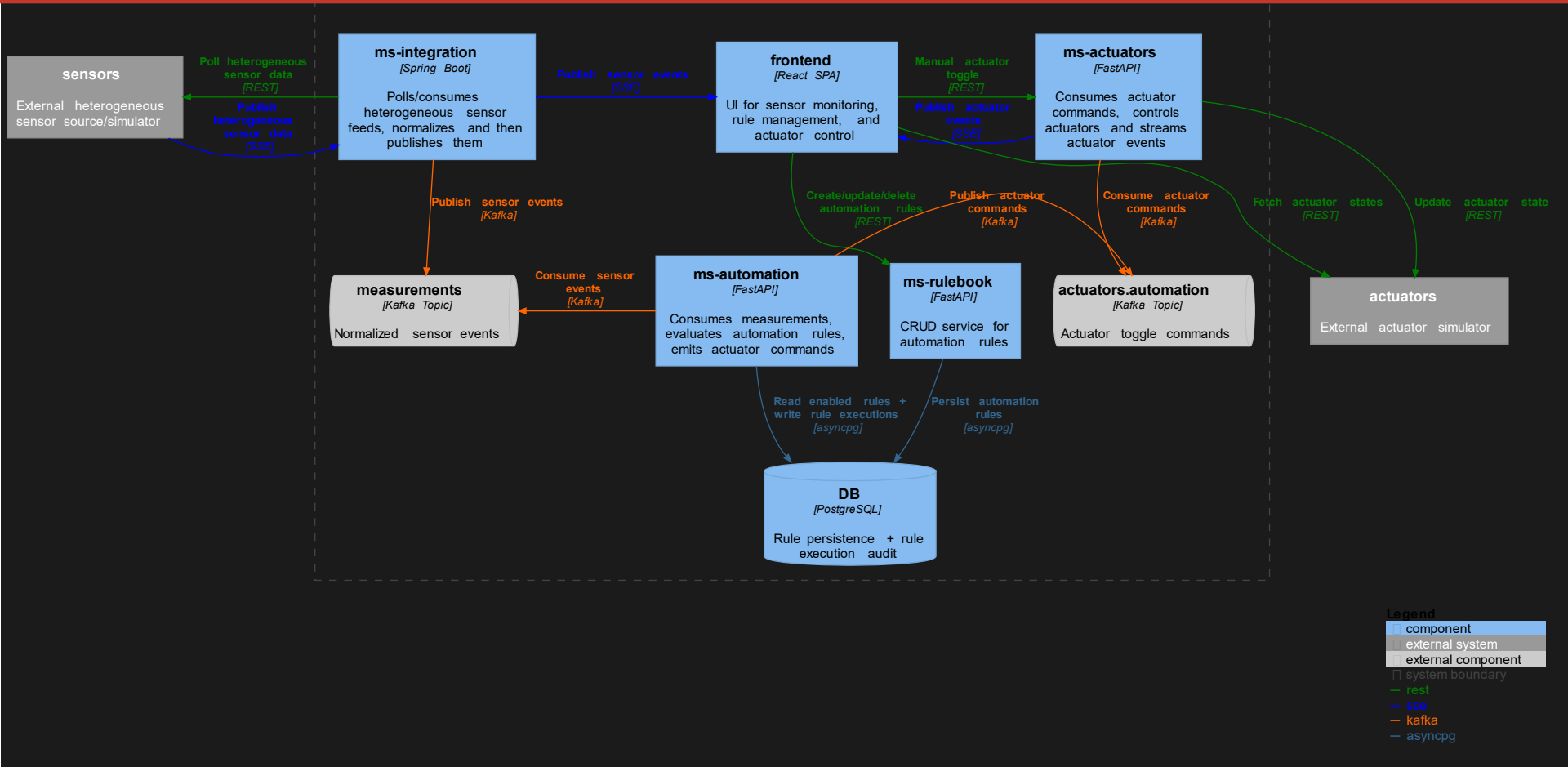
Serve a live dashboard with sensor state, actuator status, and rule management UI

System Architecture



DBMS — REST — Kafka — SSE — HTTP proxy

C4 Component Diagram



Microservices Breakdown

ms-integration :8080 — Java · Spring Boot

- Polls 8 REST sensors
- Consumes 7 SSE telemetry topics
- Publishes normalized events

ms-automation :8001 — Python · FastAPI

- Consumes Kafka sensor events
- Evaluates IF-THEN rules
- Publishes actuator commands

ms-rulebook :8003 — Python · FastAPI

- CRUD API for automation rules
- Persists rules in PostgreSQL

ms-actuators :8004 — Python · FastAPI

- Consumes actuator commands from Kafka
- Calls simulator REST API
- Exposes SSE stream of actuator events

Automation Engine (Python)

Rule Syntax

```
IF <sensor>.<metric> <op> <value>
THEN SET <actuator> TO ON|OFF
```

```
e.g.  IF greenhouse_temperature.temperature_c > 28
      THEN SET cooling_fan TO ON
```

Supported Operators

`<` `<=` `=` `>=` `>`

Evaluated on every sensor event arrival

Kafka Consumer

Consumes MeasurementEvents from the *measurements* Kafka topic

Rule Evaluation

Loads rules, matches sensor and metrics, evaluates operator & threshold

Kafka Producer

Publishes ActuatorCommands to the *actuators.automation* Kafka topic

Live Actuator Interaction

ms-actuators service applies actuactor commands and notifies the dashboard

Infrastructure & DevOps

Service	Image / Stack	Notes
kafka	apache/kafka:3.9.0	KRaft mode, single broker, auto-create topics
postgres	postgres:16-alpine	Persists automation rules; init scripts in /source/postgres
simulator	mars-iot-simulator:v1	Provided — must not be modified
ms-integration	Java · Spring Boot	Polls REST + SSE telemetry → Kafka
ms-automation	Python · FastAPI	Rule engine; consumes/produces Kafka; REST + SSE out
ms-rulebook	Python · FastAPI	CRUD rules API
ms-actuators	Python · FastAPI	Kafka consumer → simulator REST; SSE stream out
frontend	Node (React build)	SPA served on :3001 internally, proxied by Nginx
nginx	nginx:1.25-alpine	API gateway on :80; routes rules/actuators/integration/SPA

Nginx API Gateway & Routing

Nginx
:80

Single public entry point
gzip + SSE buffering config

`/api/rules/*` → `ms-rulebook:8003`	CRUD rule management
`/api/actuators/stream` → `ms-actuators:8004`	SSE stream (buffering OFF)
`/api/actuators/*` → `ms-actuators:8004`	Manual actuator control
`/api/integration/*` → `ms-integration:8080`	Sensor state & polling
`/ (catch-all)` → `frontend:3000`	React SPA + WS upgrade

Technology Stack

Backend

- Java 21 + Spring Boot
- Python 3.12 + FastAPI (×3)
- Spring Kafka (Java)
- AIOKafka (Python)

Messaging & Storage

- Apache Kafka 3.9 (KRaft)
- PostgreSQL 16
- In-memory sensor state cache
- SSE streams (native HTTP)

Frontend & Gateway

- React SPA
- nginx 1.25-alpine API gateway
- SSE for real-time updates
- REST for rule management
- Docker Compose IaC

User Stories — Dashboard & Monitoring

Operator | **Administrator** | **Automation Engineer**

View a centralized dashboard of all habitat sensors to monitor the overall health of the Mars base.

Operator

Sensor data updates automatically without refreshing so I am always looking at live, real-time telemetry.

Operator

View a live line chart for selected sensors (e.g. greenhouse temperature) while the page is open to track trends.

Operator

Dashboard displays the exact timestamp of the last data point received so I know if a sensor has gone offline.

Operator

See the most recent reading from each sensor instantly upon opening the dashboard — no waiting for the next broadcast.

Operator

See standardized units and naming conventions for all data, regardless of which device it came from.

Operator

Sensors are grouped by operational domain (Thermal, Power, Air Quality) for fast navigation during an emergency.

Operator

A visual indicator flags sensor data older than 5 minutes so I don't mistakenly trust stale telemetry.

Operator

User Stories — Automation Rule Management

Automation Engineer

Create new rules using a simple sentence format (IF sensor op value THEN actuator state).

Automation Engineer

Select sensors and actuators from predefined lists when creating a rule to avoid fatal typing errors.

Automation Engineer

Define rule conditions using standard operators (<, <=, =, >, >=) to set precise environmental thresholds.

Automation Engineer

View a list of all currently active rules to audit the habitat's full automated logic at a glance.

Automation Engineer

Delete automation rules to permanently remove outdated or dangerous logic from the system.

Automation Engineer

Edit existing rules to adjust thresholds (e.g. lower heater trigger) as Martian seasons change.

Automation Engineer

Temporarily disable and re-enable specific rules without deleting them to prevent false actuations during maintenance.

Administrator

Automation rules are permanently saved so they survive a system crash or power cycle.

User Stories — System Resilience & Configuration

Dashboard displays data from ALL sensors regardless of transmission method — polling or streaming — no blind spots.

Operator

System evaluates rules the exact instant new sensor data arrives so life-saving equipment activates immediately.

Operator

The monitoring dashboard stays live and responsive even if the automated rule system goes offline.

Operator

A visual alert appears on the UI when a sensor reads out of normal bounds — even before a rule triggers.

Operator

A global 'System Connection' indicator (Green/Red) shows immediately if the UI has lost its SSE/WebSocket connection.

Operator

Configure global polling frequency for legacy sensors to balance network load with dashboard responsiveness.

Administrator

Manually trigger a refresh of all actuator states via a dashboard button to verify true physical status.

Operator

User Story Summary

8	Dashboard & Monitoring
7	Rule Management (CRUD)
5	System Resilience
2	Config & Actuators

24 user stories implemented

Mission Accomplished.

```
docker compose up  →  http://localhost
```

Credits

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